

1. 4to. parcial

1.

$$D_x \left(\frac{x^3 - 4x^2 - 3}{-2x^5 + x^4 - 3} \right) =$$

2.

$$D_x \left(\frac{\sqrt{4x^2 + 4x - 4}}{\sqrt[5]{3x^4 + 4x^3 - 3}} \right) =$$

3.

$$\frac{d}{dx}(x + 3y)^2$$

2. 4to. parcial

1.

$$D_x \left(\frac{4x^2 - 4x + 4}{4x^3 - x^2 - 2} \right) =$$

2.

$$D_x \left(\frac{\sqrt[4]{3x^5 - 3x^4 + 2}}{\sqrt[5]{3x^4 - 2x^3 - 1}} \right) =$$

3.

$$\frac{d}{dx}(x + 3y)^2$$

3. 4to. parcial

1.

$$D_x \left(\frac{-x^3 + x^2 - 2}{2x^3 - 4x^2 - 3} \right) =$$

2.

$$D_x \left(\frac{\sqrt[5]{3x^3 - 4x^2 - 3}}{\sqrt{2x^4 + x^3 + 2}} \right) =$$

3.

$$\frac{d}{dx}(x + 4y)^4$$

4. 4to. parcial

1.

$$D_x \left(\frac{-4x^2 - x + 4}{-3x^3 + 3x^2 + 3} \right) =$$

2.

$$D_x \left(\frac{\sqrt{2x^2 - 2x - 1}}{\sqrt[3]{2x^5 + x^4 + 3}} \right) =$$

3.

$$\frac{d}{dx} (3x - 3y)^3$$

5. 4to. parcial

1.

$$D_x \left(\frac{-4x^3 - 4x^2 - 2}{3x^4 - 3x^3 - 3} \right) =$$

2.

$$D_x \left(\frac{\sqrt{2x^5 - 3x^4 + 3}}{\sqrt[4]{2x^4 + 3x^3 + 4}} \right) =$$

3.

$$\frac{d}{dx} (2x + 3y)^4$$

6. 4to. parcial

1.

$$D_x \left(\frac{x^2 + 2x + 3}{-3x^5 + 4x^4 - 3} \right) =$$

2.

$$D_x \left(\frac{\sqrt{2x^2 - 3x - 1}}{\sqrt[5]{4x^5 - 4x^4 - 3}} \right) =$$

3.

$$\frac{d}{dx}(x - 2y)^3$$

7. 4to. parcial

1.

$$D_x \left(\frac{4x^3 + x^2 + 2}{2x^4 - 3x^3 + 3} \right) =$$

2.

$$D_x \left(\frac{\sqrt[4]{3x^5 - 3x^4 - 4}}{\sqrt[5]{4x^4 + 4x^3 - 1}} \right) =$$

3.

$$\frac{d}{dx}(4x - y)^2$$

8. 4to. parcial

1.

$$D_x \left(\frac{-x^3 + 3x^2 - 3}{-x^3 - 2x^2 - 1} \right) =$$

2.

$$D_x \left(\frac{\sqrt[3]{2x^3 - 2x^2 - 1}}{\sqrt[5]{x^3 - 3x^2 - 1}} \right) =$$

3.

$$\frac{d}{dx}(4x + 3y)^4$$

9. 4to. parcial

1.

$$D_x \left(\frac{3x^2 + 3x + 2}{4x^5 - 2x^4 + 2} \right) =$$

2.

$$D_x \left(\frac{\sqrt[3]{4x^5 - 2x^4 + 4}}{\sqrt[5]{2x^2 + 4x + 4}} \right) =$$

3.

$$\frac{d}{dx}(3x - y)^2$$

10. 4to. parcial

1.

$$D_x \left(\frac{2x^3 - 2x^2 + 4}{-2x^2 + 2x + 4} \right) =$$

2.

$$D_x \left(\frac{\sqrt[3]{3x^5 + 3x^4 + 1}}{\sqrt[3]{3x^3 + 2x^2 - 1}} \right) =$$

3.

$$\frac{d}{dx} (4x + 4y)^5$$

11. 4to. parcial

1.

$$D_x \left(\frac{x^2 + x - 4}{-2x^3 + 2x^2 - 4} \right) =$$

2.

$$D_x \left(\frac{\sqrt[3]{3x^4 + 2x^3 + 2}}{\sqrt{2x^2 + 2x - 4}} \right) =$$

3.

$$\frac{d}{dx}(2x + y)^3$$

12. 4to. parcial

1.

$$D_x \left(\frac{-x^4 + 4x^3 + 1}{-x^5 - x^4 + 4} \right) =$$

2.

$$D_x \left(\frac{\sqrt[3]{3x^2 - 4x + 3}}{\sqrt[3]{2x^2 + 2x - 3}} \right) =$$

3.

$$\frac{d}{dx}(4x + 3y)^5$$

13. 4to. parcial

1.

$$D_x \left(\frac{2x^4 - 2x^3 + 2}{-2x^4 + x^3 - 1} \right) =$$

2.

$$D_x \left(\frac{\sqrt[4]{4x^3 - x^2 + 4}}{\sqrt[5]{3x^2 - 3x - 4}} \right) =$$

3.

$$\frac{d}{dx}(2x - 3y)^2$$

14. 4to. parcial

1.

$$D_x \left(\frac{-2x^5 - 3x^4 + 4}{x^2 - 3x + 2} \right) =$$

2.

$$D_x \left(\frac{\sqrt{2x^3 + x^2 - 4}}{\sqrt[4]{3x^3 + 2x^2 - 2}} \right) =$$

3.

$$\frac{d}{dx} (4x + 4y)^2$$

15. 4to. parcial

1.

$$D_x \left(\frac{4x^4 + 2x^3 + 1}{3x^3 + 3x^2 - 4} \right) =$$

2.

$$D_x \left(\frac{\sqrt{4x^5 + x^4 + 2}}{\sqrt[5]{x^3 + x^2 - 2}} \right) =$$

3.

$$\frac{d}{dx}(x - 3y)^4$$

16. 4to. parcial

1.

$$D_x \left(\frac{4x^2 - 3x + 3}{-2x^4 - 4x^3 + 3} \right) =$$

2.

$$D_x \left(\frac{\sqrt{x^4 - 2x^3 + 2}}{\sqrt[5]{x^4 - x^3 + 2}} \right) =$$

3.

$$\frac{d}{dx} (2x - 4y)^3$$

17. 4to. parcial

1.

$$D_x \left(\frac{-x^2 + 4x - 1}{-x^5 + 4x^4 - 1} \right) =$$

2.

$$D_x \left(\frac{\sqrt[4]{3x^2 - 3x - 3}}{\sqrt[3]{4x^4 + 2x^3 + 4}} \right) =$$

3.

$$\frac{d}{dx} (2x + 2y)^2$$

18. 4to. parcial

1.

$$D_x \left(\frac{4x^4 + 2x^3 - 2}{3x^3 - 2x^2 + 1} \right) =$$

2.

$$D_x \left(\frac{\sqrt{4x^4 + 3x^3 + 4}}{\sqrt[4]{2x^3 - 3x^2 + 4}} \right) =$$

3.

$$\frac{d}{dx}(2x - 2y)^5$$

19. 4to. parcial

1.

$$D_x \left(\frac{-2x^2 + 3x + 3}{-3x^5 - 4x^4 + 4} \right) =$$

2.

$$D_x \left(\frac{\sqrt{2x^3 + 2x^2 + 4}}{\sqrt[5]{x^5 + 2x^4 + 1}} \right) =$$

3.

$$\frac{d}{dx}(x + y)^4$$

20. 4to. parcial

1.

$$D_x \left(\frac{-3x^3 - 3x^2 - 2}{-x^5 - 4x^4 + 1} \right) =$$

2.

$$D_x \left(\frac{\sqrt[3]{4x^3 - 4x^2 - 1}}{\sqrt[4]{2x^2 + x - 2}} \right) =$$

3.

$$\frac{d}{dx} (4x + 4y)^4$$

21. 4to. parcial

1.

$$D_x \left(\frac{-2x^3 + x^2 - 1}{-2x^5 - 3x^4 - 3} \right) =$$

2.

$$D_x \left(\frac{\sqrt[3]{x^2 + 3x + 4}}{\sqrt[5]{x^2 + x + 2}} \right) =$$

3.

$$\frac{d}{dx} (3x + 2y)^5$$

22. 4to. parcial

1.

$$D_x \left(\frac{-3x^3 - 3x^2 + 2}{4x^3 + 2x^2 - 1} \right) =$$

2.

$$D_x \left(\frac{\sqrt[5]{3x^3 + 2x^2 - 3}}{\sqrt[4]{3x^4 - 4x^3 - 2}} \right) =$$

3.

$$\frac{d}{dx}(x - 3y)^4$$

23. 4to. parcial

1.

$$D_x \left(\frac{4x^5 + 4x^4 + 4}{4x^2 - 3x + 3} \right) =$$

2.

$$D_x \left(\frac{\sqrt[3]{x^2 - 3x - 2}}{\sqrt[3]{x^5 + 4x^4 - 3}} \right) =$$

3.

$$\frac{d}{dx} (4x + 2y)^2$$

24. 4to. parcial

1.

$$D_x \left(\frac{-3x^2 + 3x - 3}{x^4 + 3x^3 + 3} \right) =$$

2.

$$D_x \left(\frac{\sqrt[5]{2x^4 - 4x^3 + 1}}{\sqrt{x^3 - 3x^2 + 2}} \right) =$$

3.

$$\frac{d}{dx}(x + 4y)^2$$

25. 4to. parcial

1.

$$D_x \left(\frac{-3x^3 + 3x^2 - 2}{-4x^2 - 4x - 4} \right) =$$

2.

$$D_x \left(\frac{\sqrt[5]{x^3 + 3x^2 + 1}}{\sqrt[3]{4x^2 + 2x - 4}} \right) =$$

3.

$$\frac{d}{dx}(4x - 2y)^3$$

26. 4to. parcial

1.

$$D_x \left(\frac{3x^3 + 2x^2 - 4}{4x^2 + 3x - 4} \right) =$$

2.

$$D_x \left(\frac{\sqrt[3]{3x^4 + 4x^3 - 1}}{\sqrt[3]{2x^5 + 2x^4 - 1}} \right) =$$

3.

$$\frac{d}{dx}(x - 2y)^5$$

27. 4to. parcial

1.

$$D_x \left(\frac{2x^2 - 3x - 3}{-4x^5 + 3x^4 - 2} \right) =$$

2.

$$D_x \left(\frac{\sqrt[3]{x^2 + x + 1}}{\sqrt[5]{x^2 - 2x - 3}} \right) =$$

3.

$$\frac{d}{dx}(x - 3y)^5$$

28. 4to. parcial

1.

$$D_x \left(\frac{-3x^4 + 2x^3 + 1}{2x^2 + 2x + 4} \right) =$$

2.

$$D_x \left(\frac{\sqrt[5]{3x^5 - 3x^4 - 2}}{\sqrt[3]{2x^3 + 3x^2 - 1}} \right) =$$

3.

$$\frac{d}{dx} (4x + 3y)^5$$

29. 4to. parcial

1.

$$D_x \left(\frac{3x^3 - x^2 - 2}{-3x^4 - 4x^3 + 3} \right) =$$

2.

$$D_x \left(\frac{\sqrt[4]{2x^2 + 3x - 3}}{\sqrt[5]{x^4 + 2x^3 - 1}} \right) =$$

3.

$$\frac{d}{dx}(x + 2y)^2$$

30. 4to. parcial

1.

$$D_x \left(\frac{3x^4 + 4x^3 + 4}{-x^4 + x^3 - 1} \right) =$$

2.

$$D_x \left(\frac{\sqrt{3x^5 - 2x^4 - 4}}{\sqrt[4]{x^5 - x^4 - 4}} \right) =$$

3.

$$\frac{d}{dx}(2x - 3y)^2$$

31. 4to. parcial

1.

$$D_x \left(\frac{-4x^3 - 2x^2 + 4}{-2x^3 - 2x^2 - 1} \right) =$$

2.

$$D_x \left(\frac{\sqrt[4]{x^2 - x - 3}}{\sqrt[5]{x^4 - x^3 - 1}} \right) =$$

3.

$$\frac{d}{dx} (4x - 2y)^3$$

32. 4to. parcial

1.

$$D_x \left(\frac{-3x^2 + x - 1}{4x^2 - 4x - 3} \right) =$$

2.

$$D_x \left(\frac{\sqrt{3x^3 - x^2 - 2}}{\sqrt[4]{4x^5 + 2x^4 - 4}} \right) =$$

3.

$$\frac{d}{dx} (3x + 3y)^5$$

33. 4to. parcial

1.

$$D_x \left(\frac{2x^4 - 3x^3 - 3}{4x^3 - 3x^2 - 3} \right) =$$

2.

$$D_x \left(\frac{\sqrt[4]{3x^3 + x^2 + 3}}{\sqrt[5]{2x^2 + 4x - 4}} \right) =$$

3.

$$\frac{d}{dx}(2x - 3y)^4$$

34. 4to. parcial

1.

$$D_x \left(\frac{2x^5 - 3x^4 + 3}{-2x^2 + 2x + 3} \right) =$$

2.

$$D_x \left(\frac{\sqrt[5]{3x^3 - 2x^2 - 4}}{\sqrt[3]{3x^3 - 4x^2 + 3}} \right) =$$

3.

$$\frac{d}{dx} (3x + 2y)^2$$

35. 4to. parcial

1.

$$D_x \left(\frac{3x^3 - 2x^2 - 3}{-2x^3 - 2x^2 + 4} \right) =$$

2.

$$D_x \left(\frac{\sqrt{x^4 - 3x^3 - 2}}{\sqrt[5]{x^5 + 3x^4 + 1}} \right) =$$

3.

$$\frac{d}{dx}(4x + y)^4$$

36. 4to. parcial

1.

$$D_x \left(\frac{-3x^3 - 2x^2 + 3}{4x^4 - 4x^3 - 3} \right) =$$

2.

$$D_x \left(\frac{\sqrt[3]{3x^2 + 4x - 4}}{\sqrt{2x^2 - 4x + 2}} \right) =$$

3.

$$\frac{d}{dx} (2x + 3y)^2$$

37. 4to. parcial

1.

$$D_x \left(\frac{2x^2 + x - 4}{4x^2 + 4x + 4} \right) =$$

2.

$$D_x \left(\frac{\sqrt[4]{3x^5 - 4x^4 - 3}}{\sqrt[3]{4x^3 + 4x^2 + 4}} \right) =$$

3.

$$\frac{d}{dx} (2x + 3y)^2$$

38. 4to. parcial

1.

$$D_x \left(\frac{3x^2 - 3x + 1}{-2x^4 - 3x^3 + 1} \right) =$$

2.

$$D_x \left(\frac{\sqrt[4]{x^2 - x - 2}}{\sqrt[5]{2x^5 - x^4 + 3}} \right) =$$

3.

$$\frac{d}{dx}(3x + y)^4$$

39. 4to. parcial

1.

$$D_x \left(\frac{-2x^5 + x^4 - 1}{4x^2 - 3x - 1} \right) =$$

2.

$$D_x \left(\frac{\sqrt{4x^3 - x^2 + 1}}{\sqrt[3]{3x^3 - 2x^2 + 4}} \right) =$$

3.

$$\frac{d}{dx} (3x - 2y)^2$$

40. 4to. parcial

1.

$$D_x \left(\frac{-2x^3 - 3x^2 + 1}{x^5 + 2x^4 - 3} \right) =$$

2.

$$D_x \left(\frac{\sqrt[3]{3x^2 - x + 1}}{\sqrt[3]{3x^4 - 4x^3 - 1}} \right) =$$

3.

$$\frac{d}{dx} (4x - 4y)^3$$

41. 4to. parcial

1.

$$D_x \left(\frac{-4x^3 - 2x^2 - 2}{-x^3 - x^2 + 3} \right) =$$

2.

$$D_x \left(\frac{\sqrt[3]{3x^2 - x - 1}}{\sqrt[5]{2x^4 + 4x^3 + 1}} \right) =$$

3.

$$\frac{d}{dx} (3x + 4y)^5$$

42. 4to. parcial

1.

$$D_x \left(\frac{2x^3 + 3x^2 - 4}{-4x^4 + 2x^3 - 1} \right) =$$

2.

$$D_x \left(\frac{\sqrt[5]{2x^3 + x^2 - 3}}{\sqrt[5]{x^4 + 2x^3 + 1}} \right) =$$

3.

$$\frac{d}{dx} (4x + 3y)^2$$

43. 4to. parcial

1.

$$D_x \left(\frac{2x^4 + 3x^3 - 1}{3x^5 + 4x^4 - 3} \right) =$$

2.

$$D_x \left(\frac{\sqrt{2x^5 - 2x^4 + 2}}{\sqrt[4]{3x^3 + 2x^2 + 2}} \right) =$$

3.

$$\frac{d}{dx}(x - 3y)^3$$

44. 4to. parcial

1.

$$D_x \left(\frac{x^2 + x + 3}{-2x^2 + 3x - 1} \right) =$$

2.

$$D_x \left(\frac{\sqrt[4]{2x^2 - 2x - 1}}{\sqrt[4]{2x^4 - 4x^3 + 2}} \right) =$$

3.

$$\frac{d}{dx} (3x - 2y)^4$$

45. 4to. parcial

1.

$$D_x \left(\frac{3x^3 - 3x^2 - 3}{-3x^2 + 3x - 1} \right) =$$

2.

$$D_x \left(\frac{\sqrt[4]{2x^5 + 2x^4 + 3}}{\sqrt[3]{2x^3 - 4x^2 + 3}} \right) =$$

3.

$$\frac{d}{dx} (4x - 4y)^2$$